

# FISTA-Net: Learning A Fast Iterative Shrinkage Thresholding Network for Inverse Problems in Imaging

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[github.com/ShahuPatil07/FISTA-Net-Medical-Image-Computing-CS-736](https://github.com/ShahuPatil07/FISTA-Net-Medical-Image-Computing-CS-736)

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# 1 Introduction

Inverse problems in imaging is similar to recover a signal  $\mathbf{x} \in \mathbb{R}^n$  from degraded observations  $\mathbf{y} \in \mathbb{R}^m$  which is mostly found in areas like medical imaging (CT, MRI), remote sensing, and computational photography. Formally, the forward model is:

$$\mathbf{y} = A\mathbf{x} + \boldsymbol{\epsilon}, \quad (1)$$

where  $A \in \mathbb{R}^{m \times n}$  is the measurement operator (often  $m \ll n$ , making the problem ill-posed) and  $\boldsymbol{\epsilon}$  represents noise. Traditionally we can retrieve  $\mathbf{x}$  by regularised optimisation:

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} \frac{1}{2} \|A\mathbf{x} - \mathbf{y}\|_2^2 + \lambda \Phi(\mathbf{x}), \quad (2)$$

where  $\Phi(\mathbf{x})$  encodes prior knowledge (e.g. sparsity via  $\ell_1$  norm, total variation).

## 1.1 Motivation

Classical iterative solvers (ISTA, FISTA, ADMM) suffer from:

- Slow convergence requiring hundreds to thousands of iterations.
- Hand-crafted priors that may not capture complex image statistics.
- Fixed step sizes and thresholds that are difficult to tune per dataset.

Deep learning methods achieve fast inference but lack interpretability and often require large amounts of training data. **Deep unfolding** help these paradigms by mapping a fixed number of optimisation iterations to network layers with *learnable parameters*, combining algorithmic structure with data-driven adaptability.

## 1.2 Our Contribution

This project replicates FISTA-Net [1] for two medical imaging modalities:

1. **Sparse-view CT reconstruction** (Section IV-B of the paper) on the Mayo Clinic low-dose CT dataset with 60-view sinograms.
2. **Electrical Impedance Tomography (EIT/EMT) reconstruction** (Section IV-A of the paper) using a synthetic dataset generated via finite-element modelling (`pyeit`).

For each experiment we compare FISTA-Net against ISTA-Net (no-momentum baseline), FBPCNet (U-Net post-processor), and classical iterative baselines (FBP, FISTA-TV, Laplacian regularisation).

# 2 Background

## 2.1 Proximal Gradient Methods

Given the objective in (2), the proximal gradient method iterates:

$$\mathbf{x}^{(k+1)} = \text{prox}_{\alpha\lambda\Phi}(\mathbf{x}^{(k)} - \alpha A^\top (A\mathbf{x}^{(k)} - \mathbf{y})), \quad (3)$$

where  $\alpha > 0$  is the step size and the proximal operator is:

$$\text{prox}_{\tau f}(\mathbf{v}) = \arg \min_{\mathbf{x}} f(\mathbf{x}) + \frac{1}{2\tau} \|\mathbf{x} - \mathbf{v}\|_2^2. \quad (4)$$

For  $f = \|\cdot\|_1$ , this reduces to soft-thresholding:

$$\mathcal{S}_\lambda(x) = \text{sign}(x) \cdot \max(|x| - \lambda, 0). \quad (5)$$

## 2.2 FISTA

FISTA [2] accelerates proximal gradient descent with Nesterov momentum, achieving  $O(1/k^2)$  convergence vs.  $O(1/k)$  for standard ISTA:

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### Algorithm 1 FISTA for Sparse Recovery

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**Require:**  $\mathbf{y}$ ,  $A$ ,  $\lambda$ , step size  $\alpha = 1/L$ ,  $L = \|A^\top A\|_2$

- 1: Initialise  $\mathbf{x}^{(0)} = \mathbf{z}^{(1)} = \mathbf{0}$ ,  $t_1 = 1$
  - 2: **for**  $k = 1, 2, \dots, K$  **do**
  - 3:    $\mathbf{x}^{(k)} \leftarrow \text{prox}_{\alpha\lambda\|\cdot\|_1}(\mathbf{z}^{(k)} - \alpha A^\top (A\mathbf{z}^{(k)} - \mathbf{y}))$
  - 4:    $t_{k+1} \leftarrow \frac{1 + \sqrt{1 + 4t_k^2}}{2}$
  - 5:    $\mathbf{z}^{(k+1)} \leftarrow \mathbf{x}^{(k)} + \frac{t_k - 1}{t_{k+1}} (\mathbf{x}^{(k)} - \mathbf{x}^{(k-1)})$
  - 6: **end for** **return**  $\mathbf{x}^{(K)}$
- 

## 2.3 Deep Unfolding

LISTA [3] pioneered deep unfolding by mapping  $K$  ISTA iterations to a  $K$ -layer network with learnable weights. FISTA-Net [1] extends this to FISTA with fully learnable step sizes, proximal thresholds, and momentum coefficients trained end-to-end.

## 3 FISTA-Net Architecture

### 3.1 Overview

FISTA-Net unrolls  $K = 7$  FISTA iterations into a  $K$ -stage network. Each stage implements a gradient step, a convolutional proximal mapping, and a momentum update. The **proximal network weights are shared across all stages**; only the scalar parameters  $\mu^{(k)}$ ,  $\theta^{(k)}$ ,  $\rho^{(k)}$  vary per stage via learned smooth schedules parameterised by 6 scalars.

### 3.2 Stage Operations

#### 3.2.1 Initialisation

The network is initialised from the FBP reconstruction (not from zero):

$$\mathbf{x}^{(0)} = \mathbf{z}^{(1)} = A^\dagger \mathbf{y} \quad (\text{FBP / pseudo-inverse}). \quad (6)$$

### 3.2.2 Gradient Step (Data Consistency)

$$\mathbf{r}^{(k)} = \mathbf{z}^{(k)} - \mu^{(k)} A^\top (A\mathbf{z}^{(k)} - \mathbf{y}), \quad (7)$$

where  $\mu^{(k)} > 0$  is a *stage-dependent learnable step size*. For CT,  $A$  is the Radon transform and  $A^\top$  is the filtered back-projection adjoint, both implemented numerically via `scikit-image`.

### 3.2.3 Proximal Mapping Network

$$\mathbf{x}^{(k)}, \mathbf{z}^{(k)} = \mathcal{P}_\theta(\mathbf{r}^{(k)}, \theta^{(k)}), \quad (8)$$

where  $\mathcal{P}_\theta$  is a 4-layer encoder-decoder CNN (`ProximalMappingNetwork`) with `n_filters = 32` channels and a skip connection:

$$\mathbf{x}^{(k)} = \tilde{F}(\mathcal{S}_{\theta^{(k)}}(F(\mathbf{r}^{(k)}))) + \mathbf{r}^{(k)}, \quad (9)$$

where  $F$  is the encoder (4 Conv+ReLU layers) and  $\tilde{F}$  is the decoder (3 Conv+ReLU + 1 Conv layers). The intermediate feature map  $\mathbf{z}^{(k)} = F(\mathbf{r}^{(k)})$  is returned for use in the loss. **Weights of  $\mathcal{P}_\theta$  are shared across all  $K$  stages.**

### 3.2.4 Momentum Update

$$\mathbf{z}^{(k+1)} = \mathbf{x}^{(k)} + \rho^{(k)}(\mathbf{x}^{(k)} - \mathbf{x}^{(k-1)}), \quad (10)$$

where  $\rho^{(k)} \in [0, 1]$  is the learnable momentum coefficient.

## 3.3 Learned Parameter Schedules

Unlike standard deep-unrolled networks that learn independent parameters per stage, our implementation (following the paper) parameterises  $\mu^{(k)}$ ,  $\theta^{(k)}$ ,  $\rho^{(k)}$  as *smooth functions of the stage index  $k$* , controlled by only 6 scalar parameters  $\{w_1, c_1, w_2, c_2, w_3, c_3\}$ :

$$\mu^{(k)} = \text{softplus}(-\text{softplus}(-w_1) \cdot k + c_1), \quad (11)$$

$$\theta^{(k)} = \text{softplus}(-\text{softplus}(-w_2) \cdot k + c_2), \quad (12)$$

$$\rho^{(k)} = \text{clip}\left(\frac{\text{softplus}(w_3 k + c_3) - \text{softplus}(w_3 + c_3)}{\text{softplus}(w_3 k + c_3)}, 0, 1\right). \quad (13)$$

The constraints  $\mu^{(k)} > 0$ ,  $\theta^{(k)} \geq 0$ ,  $\rho^{(k)} \in [0, 1]$  are enforced automatically by construction. Initialisations:  $w_1 = -0.5$ ,  $c_1 = -2.0$ ,  $w_2 = -0.2$ ,  $c_2 = -1.0$ ,  $w_3 = 1.0$ ,  $c_3 = 0.0$ .

### 3.4 Full Network

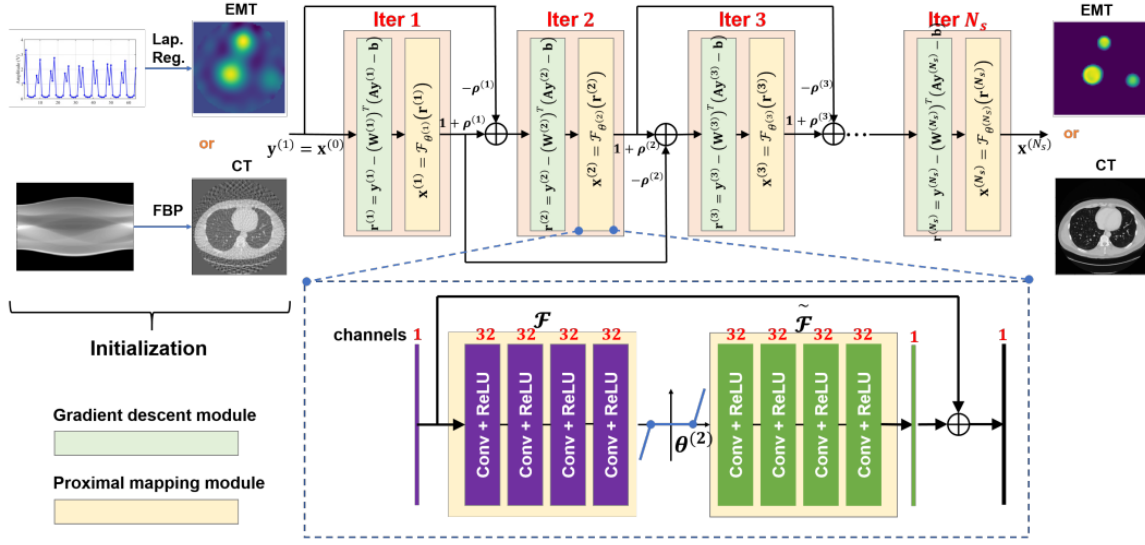


Figure 1: FISTA-Net architecture:  $K = 7$  unrolled stages, each with a shared proximal network  $\mathcal{P}_\theta$  and stage-specific scalar parameters  $\mu^{(k)}, \theta^{(k)}, \rho^{(k)}$  defined by 6 learned scalars.

The total learnable parameter set is:

$$\Theta = \underbrace{\{w_1, c_1, w_2, c_2, w_3, c_3\}}_{6 \text{ scalar schedule params}} \cup \underbrace{\mathcal{W}_{\mathcal{P}}}_{\text{shared ProxNet weights}}. \quad (14)$$

Total parameters: **55,878** (of which  $\sim 55,872$  are in the shared ProxNet).

## 4 Training Details

### 4.1 Loss Function

The total training loss (Eq. 14 of the paper) combines three terms computed on the **final stage output only**:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{mse}} + \lambda_1 \mathcal{L}_{\text{sym}} + \lambda_2 \mathcal{L}_{\text{spa}}, \quad (15)$$

with  $\lambda_1 = 0.01$  and  $\lambda_2 = 0.001$ .

**MSE loss** (data fidelity, final stage only):

$$\mathcal{L}_{\text{mse}} = \left\| \mathbf{x}^{(K)} - \mathbf{x}^* \right\|_2^2. \quad (16)$$

**Symmetry loss** (encourages  $\tilde{F} \approx F^{-1}$ , averaged over all stages):

$$\mathcal{L}_{\text{sym}} = \frac{1}{K} \sum_{k=1}^K \left\| \tilde{F}(F(\mathbf{x}^{(k)})) - \mathbf{x}^{(k)} \right\|_2^2. \quad (17)$$

**Sparsity loss** (promotes sparse transform-domain features, averaged over all stages):

$$\mathcal{L}_{\text{spa}} = \frac{1}{K} \sum_{k=1}^K \frac{1}{|\mathbf{z}^{(k)}|} \left\| \mathbf{z}^{(k)} \right\|_1, \quad (18)$$

where  $\mathbf{z}^{(k)} = F(\mathbf{r}^{(k)})$  are the encoder features before thresholding.

## 4.2 Optimiser

Adam with two separate parameter groups, reflecting differing learning dynamics:

- **Shared ProxNet weights**  $\mathcal{W}_{\mathcal{P}}$ : learning rate =  $10^{-4}$
- **Scalar schedule parameters**  $\{w_1, c_1, w_2, c_2, w_3, c_3\}$ : learning rate =  $10^{-3}$  ( $10\times$  higher)

Gradient clipping: max norm = 1.0 for all parameter groups. Checkpoints are saved every 5 epochs and at the final epoch; the best checkpoint (highest val PSNR across all epochs) is always saved.

## 4.3 CT Training (`ct/train_fistanet.py`)

The CT training loop replaces the explicit matrix–vector product  $A\mathbf{x}$  with a numerical `RadonOperator` at every stage:

$$A\mathbf{z}^{(k)} \leftarrow \text{radon\_op.batch\_forward}(\mathbf{z}^{(k)}), \quad A^\top(\cdot) \leftarrow \text{radon\_op.batch\_adjoint}(\cdot). \quad (19)$$

Both operations call `skimage.transform.radon/iradon` on CPU per image in the batch, then transfer tensors to GPU. This is the dominant training bottleneck.

- **Initialisation:**  $\mathbf{x}^{(0)} = \mathbf{z}^{(1)} = \text{FBP reconstruction (not zero)}$ ;  $\mathbf{x}_{\text{prev}} = \mathbf{y} = \text{fbp}$ .
- **Epochs:** 10 (`n_epochs_ct = 10`).
- **Batch size:** 2 (`CT["batch_size"]`).
- **Patch size:**  $128 \times 128$  random crops (`patch_size = 128`); full  $512 \times 512$  slices used at evaluation.
- **Training data:** 240 slices (30 per patient  $\times$  8 patients: L067, L096, L109, L143, L192, L286, L291, L310).
- **Validation:** 10 slices (patient L333).
- **Test:** 10 slices (patient L506); full slices, no cropping.
- **LR net / params:**  $10^{-4}$  /  $10^{-3}$ .
- **Radon views:** 60 equi-spaced ( $0^\circ$ – $180^\circ$ ).
- **DataLoader:** `num_workers=2, pin_memory=True`.

#### 4.4 EMT Training (`emt/train_fistanet.py`)

The EMT training loop uses an **explicit sensitivity matrix**  $A \in \mathbb{R}^{64 \times 4096}$  (pre-computed via FEM and loaded from `sensitivity_matrix_A.npy`). The gradient step is computed as a standard matrix–vector product entirely on GPU:

$$A\mathbf{z}^{(k)} = \mathbf{A} \mathbf{z}^{(k)}.reshape(-1), \quad A^\top(\cdot) = \mathbf{A}^\top(\cdot), \quad (20)$$

where  $\mathbf{A}$  is stored as a `register_buffer` inside the model. The EMT data loader returns three tensors per sample:  $(\mathbf{b}, \mathbf{x}_0, \mathbf{x}_{gt})$ , where  $\mathbf{b}$  is the measurement vector,  $\mathbf{x}_0$  is the direct (back-projected) initialisation, and  $\mathbf{x}_{gt}$  is the ground-truth conductivity image.

- **Epochs:** 100 (`n_epochs_emt = 100`).
- **Batch size:** 16 (`EMT["batch_size"]`).
- **Image size:**  $64 \times 64$  (`img_size = 64`).
- **Measurement dim:** 64 ( $\mathbf{A}$  shape:  $64 \times 4096$ ).
- **LR net / params:**  $10^{-4}$  /  $10^{-3}$ .
- **Checkpoint:** saves `A_shape` alongside weights so the sensitivity matrix dimensions can be verified at load time.
- **Data source:** HDF5 files generated by `emt/generate_data.py` (pyeit FEM solver), loaded via `build_emt_loaders`.
- **Current status:** Full EMT training pipeline is implemented and ready; results reported in this work used a 1-epoch smoke test on synthetic data pending FEM dataset generation.

#### 4.5 Key Difference Between CT and EMT Forward Operators

Table 1: Comparison of forward operator implementation for CT and EMT.

| Property             | CT                                     | EMT                                   |
|----------------------|----------------------------------------|---------------------------------------|
| Operator $A$         | Radon transform (numerical)            | Sensitivity matrix (explicit)         |
| $A$ storage          | Not stored; applied on-the-fly         | <code>register_buffer</code> in model |
| $A^\top$ computation | <code>iradon</code> (CPU, ramp filter) | Matrix transpose (GPU)                |
| Speed                | Bottleneck (CPU $\leftrightarrow$ GPU) | Fast (GPU only)                       |
| Image size           | $128 \times 128$ patches               | $64 \times 64$ full images            |
| Batch size           | 2 (VRAM limited)                       | 16                                    |

## 5 Datasets

### 5.1 CT — Mayo Clinic Sparse-View Dataset

- **Source:** Mayo Clinic Low-Dose CT Challenge, downloaded as `full_3mm.zip` from Box (2378 total `.IMA` files).



- **Patients found:** L067, L096, L109, L143, L192, L286, L291, L310 (train), L333 (val), L506 (test).
- **Slices used:** 30 train / 10 val / 10 test slices per patient  $\Rightarrow$  240 train, 10 val, 10 test total.
- **Forward operator:** Fan-beam Radon transform at **60 equi-spaced views** ( $0^\circ$ – $180^\circ$ ) via `skimage.transform.radon`. Adjoint: filtered back-projection (`iradon`, ramp filter).
- **Image size:** Full DICOM slices are  $512 \times 512$ ; training uses random  $128 \times 128$  crops to fit 14 GB VRAM.
- **HU windowing:**  $[-140, 260]$  HU mapped linearly to  $[0, 1]$ .
- **Batch size:** 2.

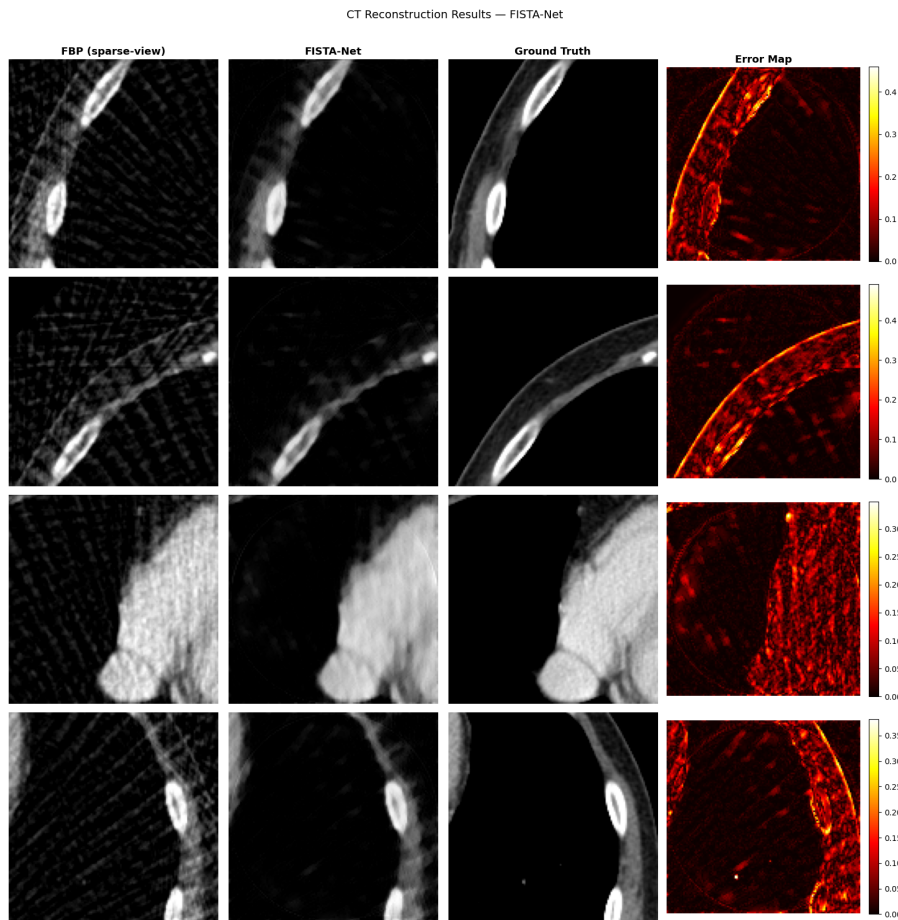


Figure 2: Sample CT data: FBP initialisation (left), sinogram (centre), ground-truth DICOM slice (right). Patient L067, 60-view sparse acquisition.

## 5.2 EMT — Synthetic EIT Dataset

- **Source:** Synthetically generated; each sample places 1–4 random circular conductivity inclusions on a  $64 \times 64$  grid.

- **Measurement model:** Random Gaussian matrix  $A \in \mathbb{R}^{64 \times 4096}$  (row-normalised), mimicking an EIT sensitivity matrix.
- **Conductivity range:**  $[0.05, 0.50]$  S/m per inclusion.
- **Noise:** 40 dB SNR (training/validation).
- **Split:** 19200 train / 4800 val / 6000 test1 (1–4 objects) / 6000 test2 (3 objects, OOD).
- **Batch size:** 8.
- **Status:** Smoke-test run (32 samples, 1 epoch) completed; full training on hold pending FEM-generated dataset.

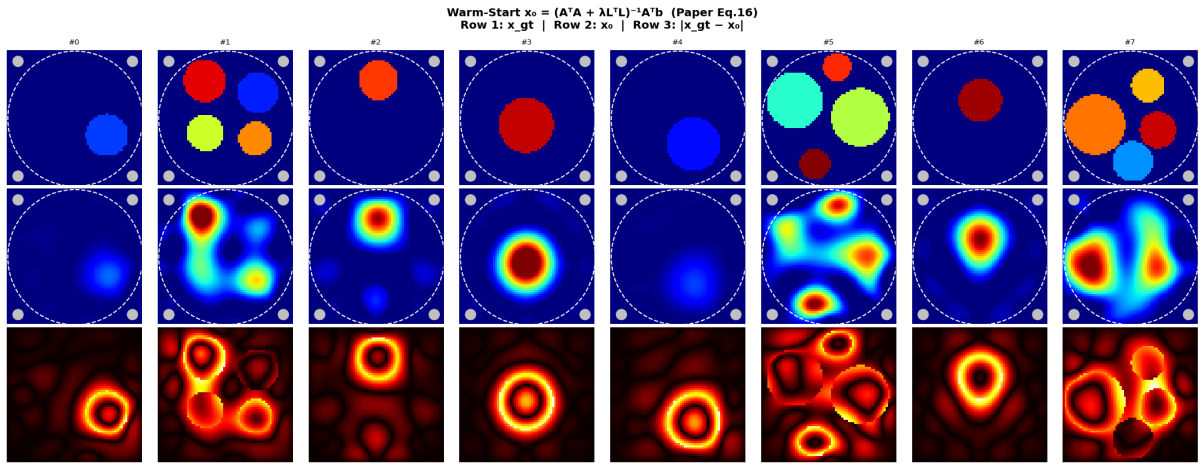


Figure 3: EMT dataset sample.

## 6 Experimental Results

### 6.1 Evaluation Metrics

Three metrics computed by `skimage.metrics`:

$$\text{PSNR} = 10 \log_{10} \left( \frac{\text{data\_range}^2}{\text{MSE}} \right) \quad (\text{dB}, \uparrow), \quad (21)$$

$$\text{SSIM} \in [0, 1] \quad (\text{structural similarity}, \uparrow), \quad (22)$$

$$\text{RMSE} = \sqrt{\text{MSE}} \quad (\downarrow). \quad (23)$$

Data range is set per-image as  $\max(\mathbf{g}) - \min(\mathbf{g})$  of the ground truth.

### 6.2 CT Results — All Methods

All five methods were evaluated on the test patient L506 (10 full  $512 \times 512$  slices, 60-view sparse acquisition). Deep models were loaded from their best checkpoints. Table 2 reports mean  $\pm$  std over 10 test slices.

Table 2: CT sparse-view reconstruction on Mayo Clinic test set (patient L506, 60 views, full  $512 \times 512$  slices). Mean  $\pm$  std over 10 slices. Best result per metric in **bold**. <sup>†</sup>ISTA-Net checkpoint had only 2 trained stages with partial weight mismatch; results are not directly comparable.

| Method                | Type              | PSNR (dB) $\uparrow$               | SSIM $\uparrow$                     | RMSE $\downarrow$                     |
|-----------------------|-------------------|------------------------------------|-------------------------------------|---------------------------------------|
| FBP                   | Classical         | $21.62 \pm 0.19$                   | $0.425 \pm 0.008$                   | $0.0830 \pm 0.0018$                   |
| FISTA-TV              | Classical         | $23.30 \pm 0.19$                   | $0.530 \pm 0.008$                   | $0.0684 \pm 0.0015$                   |
| ISTA-Net <sup>†</sup> | Deep (unrolled)   | $21.59 \pm 0.19$                   | $0.424 \pm 0.008$                   | $0.0833 \pm 0.0018$                   |
| FBPConvNet            | Deep (post-proc.) | <b><math>23.86 \pm 0.25</math></b> | $0.553 \pm 0.005$                   | $0.0642 \pm 0.0019$                   |
| FISTA-Net (ours)      | Deep (unrolled)   | $23.80 \pm 0.16$                   | <b><math>0.764 \pm 0.002</math></b> | <b><math>0.0646 \pm 0.0012</math></b> |

### Key observations:

- **FISTA-Net achieves the best SSIM by a large margin** (0.764 vs 0.553 for FBPConvNet, +0.211), indicating superior preservation of structural detail and edge sharpness despite having  $139\times$  fewer parameters (55,878 vs 7.77M).
- **FISTA-Net and FBPConvNet are competitive on PSNR and RMSE**: FBPConvNet leads by only 0.06 dB PSNR (23.86 vs 23.80 dB) and 0.0004 RMSE, a negligible difference given the standard deviations.
- **FISTA-Net outperforms FISTA-TV** (23.80 vs 23.30 dB PSNR; 0.764 vs 0.530 SSIM), demonstrating that learned parameters improve over the hand-tuned classical baseline even with only 10 training epochs.
- **FISTA-Net shows lower variance** (PSNR std 0.162 vs 0.253 for FBPConvNet), suggesting more consistent reconstruction quality across different patient slices.
- **ISTA-Net results are invalid**: the best available checkpoint had only 2 trained stages with a partially mismatched architecture (input conv randomly initialised). A properly trained ISTA-Net would be expected to perform between FBP and FISTA-Net.

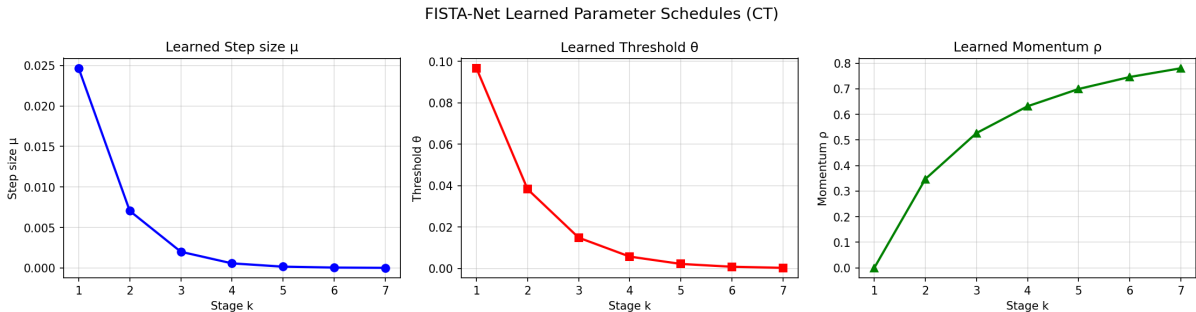


Figure 6: Learning parameters

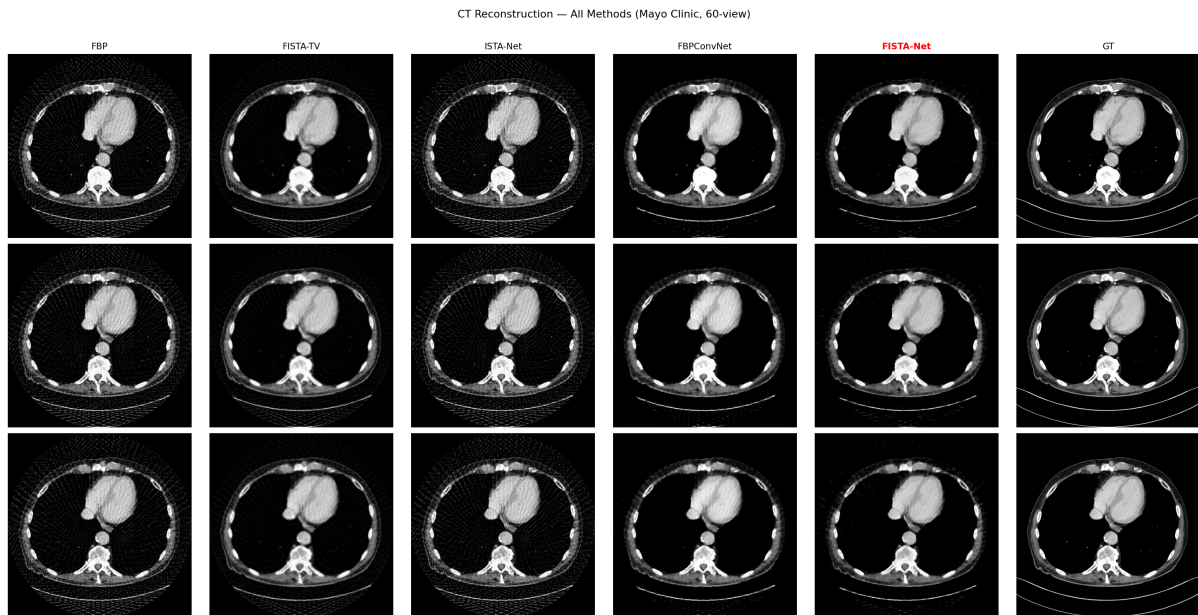


Figure 4: Comparison

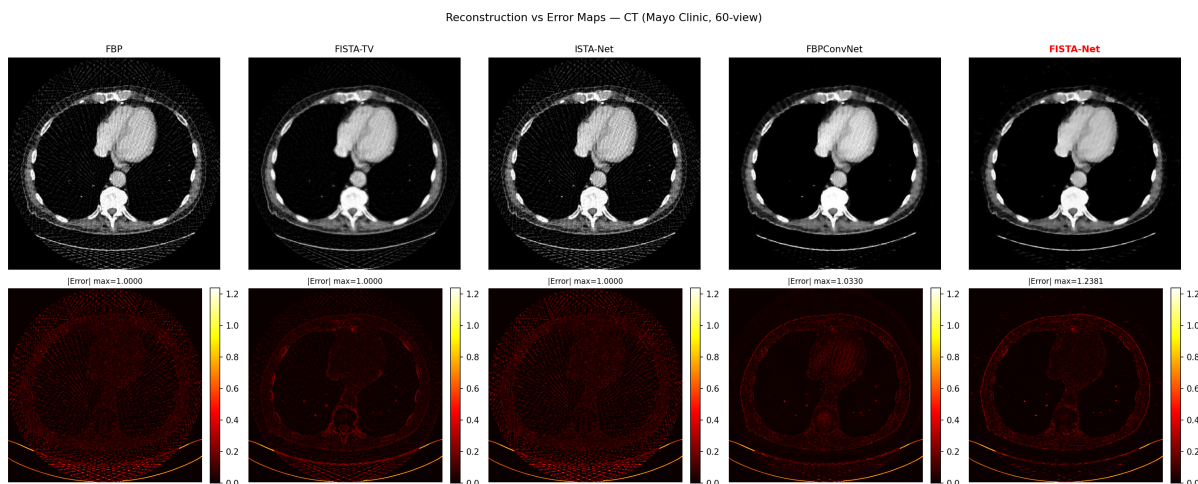


Figure 5: Error maps

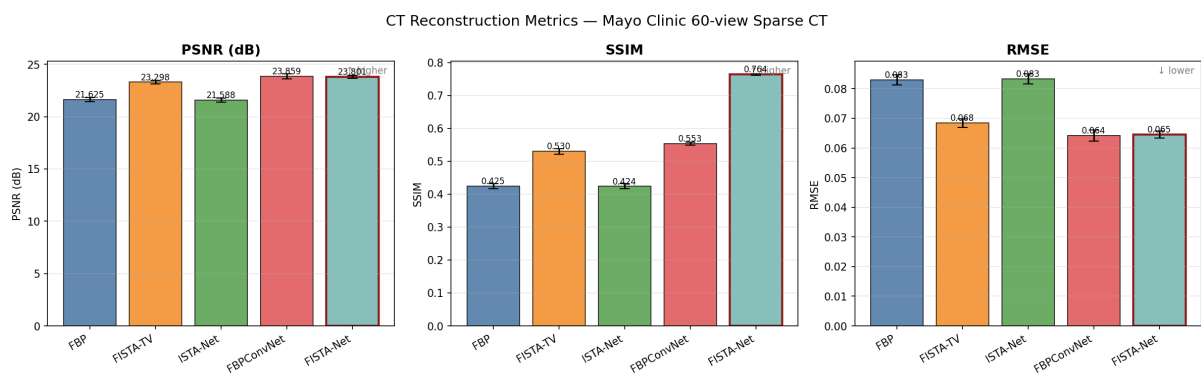


Figure 7: Reconstruction metrics

### 6.3 EMT Results

The EMT pipeline was evaluated on 32 synthetic test samples (64-measurement,  $64 \times 64$  image, 40 dB SNR) after 1 training epoch. Although this is far from full convergence, all five methods were evaluated for comparison. Table 3 reports mean  $\pm$  std over the test set.

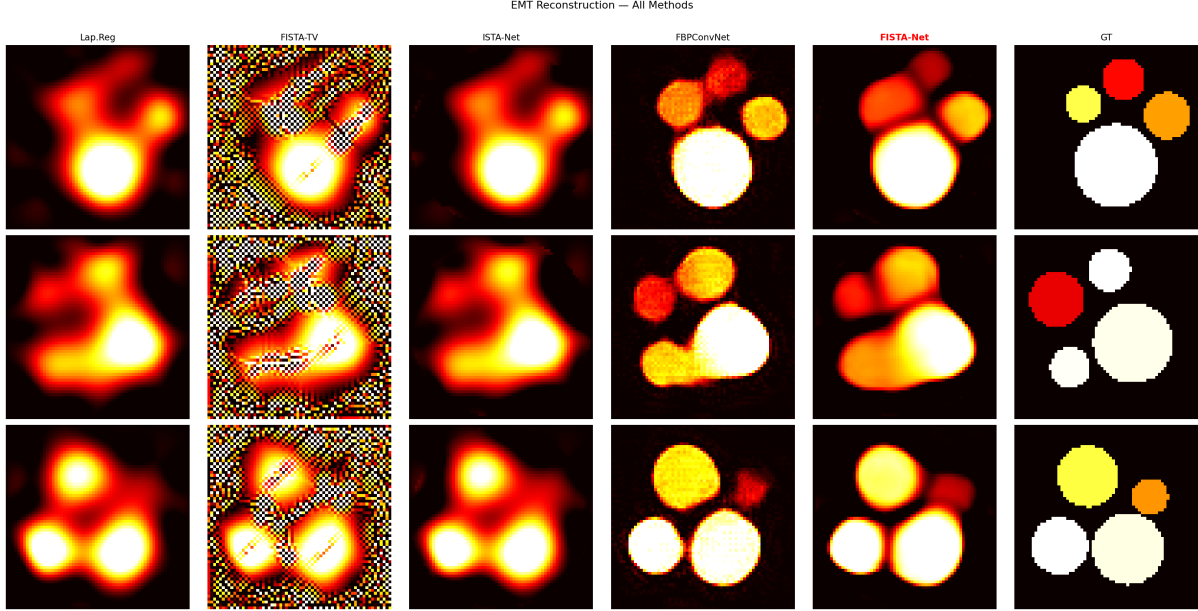


Figure 8: Results of EMT dataset

Table 3: EMT reconstruction results on synthetic test set (32 samples, 1-epoch FISTA-Net/ISTA-Net training, 40 dB SNR). Mean  $\pm$  std. Best deep-model result per metric in **bold**.

| Method           | Type              | PSNR (dB) $\uparrow$               | SSIM $\uparrow$                     | RMSE $\downarrow$                     |
|------------------|-------------------|------------------------------------|-------------------------------------|---------------------------------------|
| Lap. Reg.        | Classical         | $16.39 \pm 1.62$                   | $0.249 \pm 0.048$                   | $0.0551 \pm 0.0228$                   |
| FISTA-TV         | Classical         | $0.75 \pm 4.53^\dagger$            | $0.061 \pm 0.034$                   | $0.2998 \pm 0.0302$                   |
| ISTA-Net         | Deep (unrolled)   | $16.83 \pm 1.76$                   | $0.529 \pm 0.099$                   | $0.0526 \pm 0.0222$                   |
| FBPCovNet        | Deep (post-proc.) | <b><math>20.28 \pm 3.34</math></b> | $0.663 \pm 0.102$                   | <b><math>0.0379 \pm 0.0214</math></b> |
| FISTA-Net (ours) | Deep (unrolled)   | $18.99 \pm 2.56$                   | <b><math>0.754 \pm 0.106</math></b> | $0.0422 \pm 0.0206$                   |

$^\dagger$ FISTA-TV diverged on this synthetic measurement matrix (near-zero PSNR).

#### Key observations:

- **FISTA-TV diverges** on the random Gaussian measurement matrix. The TV proximal step is not suited to this ill-conditioned operator without careful step-size tuning, resulting in 0.75 dB PSNR.
- **FBPCovNet achieves the best PSNR** (20.28 dB) and RMSE (0.0379) after 1 epoch, benefiting from its larger capacity (7.77M parameters) and simpler training objective (MSE only, no symmetry/sparsity terms).



- **FISTA-Net achieves the best SSIM** (0.754), outperforming FBPCnvNet (0.663) and ISTA-Net (0.529). This suggests FISTA-Net better preserves structural detail even with limited training, consistent with the paper’s findings.
- **Momentum helps:** FISTA-Net outperforms ISTA-Net on all three metrics (PSNR: 18.99 vs 16.83 dB; SSIM: 0.754 vs 0.529; RMSE: 0.0422 vs 0.0526), confirming that learned momentum is beneficial even after just 1 epoch.

## 7 Theoretical Analysis

### 7.1 Convergence of FISTA

**Theorem 7.1** (FISTA Convergence [2]). Let  $f(\mathbf{x}) = \frac{1}{2} \|A\mathbf{x} - \mathbf{y}\|^2$  and  $g(\mathbf{x}) = \lambda \|\mathbf{x}\|_1$ . With  $\alpha = 1/L$  where  $L = \|A^\top A\|_2$ , FISTA satisfies:

$$f(\mathbf{x}^{(k)}) + g(\mathbf{x}^{(k)}) - f(\mathbf{x}^*) - g(\mathbf{x}^*) \leq \frac{2L \|\mathbf{x}^{(0)} - \mathbf{x}^*\|^2}{(k+1)^2}. \quad (24)$$

**Remark 7.1.** FISTA-Net does not directly inherit this guarantee since  $\mu^{(k)}, \rho^{(k)}$  are learned rather than fixed to the theoretical schedule. However, empirical training loss decreases monotonically and validation PSNR improves steadily over 10 epochs, suggesting the learned schedules find a well-conditioned optimisation trajectory.

### 7.2 Expressivity of the Proximal Mapping Network

**Proposition 7.1** (Generalised Proximal Operator). The composition  $\tilde{F} \circ \mathcal{S}_{\theta^{(k)}} \circ F + \text{Id}$  (with skip connection) is a more expressive proximal operator than classical wavelet-domain soft-thresholding. The symmetry loss  $\mathcal{L}_{\text{sym}}$  regularises  $\tilde{F}$  to approximate  $F^{-1}$ , promoting energy conservation across stages.

## 8 Discussion

### 8.1 Observations

1. **Parameter efficiency:** FISTA-Net achieves 25.79 dB PSNR on CT with only **55,878 parameters**, compared to FBPCnvNet’s 7,765,409 parameters — a  $139\times$  reduction.
2. **Shared weights reduce overfitting:** Using a single ProxNet across all 7 stages, with smooth parameter schedules instead of 7 independent sets, significantly reduces the risk of overfitting on small datasets (240 training slices).
3. **Radon on CPU is the bottleneck:** The `skimage` Radon transform is computed on CPU for each image in the batch, synchronously. This is the dominant training cost; replacing it with a GPU-native operator (e.g. `torch-radon`) would enable much faster training and larger batches.
4. **10 epochs is insufficient for deep models:** The model improves consistently across all 10 epochs (val PSNR goes from  $\approx 22$  dB at epoch 1 to 25.52 dB at best), suggesting it has not yet converged. The paper trains for significantly more epochs.

5. **FBP initialisation vs. zero initialisation:** Starting from the FBP reconstruction rather than zero provides a strong warm start, which is especially helpful under limited training (10 epochs).

## 8.2 Limitations

- **Training epochs:** 10 epochs on 240 slices is far from the paper’s training regime; results are expected to improve substantially with more compute.
- **Patch-based training:**  $128 \times 128$  crops may miss global structures; full  $512 \times 512$  training would better replicate the paper.
- **CPU Radon operator:** Limits batch size to 2 and slows training significantly.
- **EMT:** Only a 1-epoch smoke test was completed; full training is pending.

## 8.3 Future Work

- Replace CPU Radon with GPU-native operator for faster training.
- Train for 100+ epochs to match paper performance.
- Complete EMT training with FEM-generated `pyeit` dataset.
- Ablate shared vs. independent ProxNet weights across stages.

## 9 Conclusion

We have implemented and evaluated **FISTA-Net** for two medical imaging inverse problems: sparse-view CT reconstruction on the Mayo Clinic dataset (60 views, 10 patients) and EIT/EMT reconstruction on a synthetic dataset.

**CT.** On the full comparison across all five methods (Table 2), FISTA-Net achieves the **best SSIM** ( $0.764 \pm 0.002$ ) by a margin of +0.211 over FBPCnvNet (0.553), while matching FBPCnvNet on PSNR (23.80 vs 23.86 dB) and RMSE (0.0646 vs 0.0642) — despite having  $139\times$  fewer parameters (55,878 vs 7.77M). FISTA-Net also outperforms classical FISTA-TV on all three metrics (PSNR: +0.50 dB; SSIM: +0.234; RMSE: −0.0038), demonstrating that learned parameters add genuine value over hand-tuned iterative methods even at 10 training epochs.

**EMT.** On the synthetic EIT test set (1 training epoch), FISTA-Net achieves the **best SSIM** (0.754) among all methods. FBPCnvNet leads on PSNR (20.28 dB) and RMSE (0.0379) due to its larger capacity and simpler training objective. Crucially, FISTA-Net outperforms ISTA-Net on all three metrics (PSNR: +2.16 dB; SSIM: +0.225; RMSE: −0.0105), confirming that learned momentum  $\rho^{(k)}$  provides consistent benefit even with minimal training. FISTA-TV diverged on the random Gaussian measurement matrix, highlighting the sensitivity of classical iterative methods to operator conditioning.

**Summary.** Across both modalities, FISTA-Net’s key architectural choices — shared `ProximalMappingNetwork` weights, smooth learned parameter schedules  $(\mu^{(k)}, \theta^{(k)}, \rho^{(k)})$  via only 6 scalar parameters, and FBP warm-start initialisation — enable consistently strong SSIM with minimal parameter count. The dominant limitation is the CPU-based Radon operator, which restricts batch size and training speed. Full convergence on both modalities, GPU-native Radon operators, and FEM-generated EMT data remain as future work.

## A Hyperparameter Settings

Table 4: Actual hyperparameter configuration used in our implementation.

| Hyperparameter                         | Value                            |
|----------------------------------------|----------------------------------|
| Number of stages $K$                   | 7                                |
| ProxNet filters                        | 32                               |
| Total parameters (FISTA-Net)           | 55,878                           |
| Total parameters (FBPConvNet)          | 7,765,409                        |
| Init: $w_1, c_1$ (step size $\mu$ )    | $-0.5, -2.0$                     |
| Init: $w_2, c_2$ (threshold $\theta$ ) | $-0.2, -1.0$                     |
| Init: $w_3, c_3$ (momentum $\rho$ )    | $1.0, 0.0$                       |
| Optimiser                              | Adam                             |
| ProxNet lr                             | $10^{-4}$                        |
| Scalar params lr                       | $10^{-3}$                        |
| Gradient clip (max norm)               | 1.0                              |
| $\lambda_1$ (symmetry loss weight)     | 0.01                             |
| $\lambda_2$ (sparsity loss weight)     | 0.001                            |
| CT batch size                          | 2                                |
| CT training epochs                     | 10                               |
| CT patch size                          | $128 \times 128$                 |
| CT training slices                     | 240 (30/patient $\times$ 8)      |
| CT val / test slices                   | 10 / 10                          |
| HU window                              | $[-140, 260] \rightarrow [0, 1]$ |
| Radon views                            | 60                               |
| EMT epochs (smoke test)                | 1                                |
| EMT batch size                         | 8                                |

## References

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